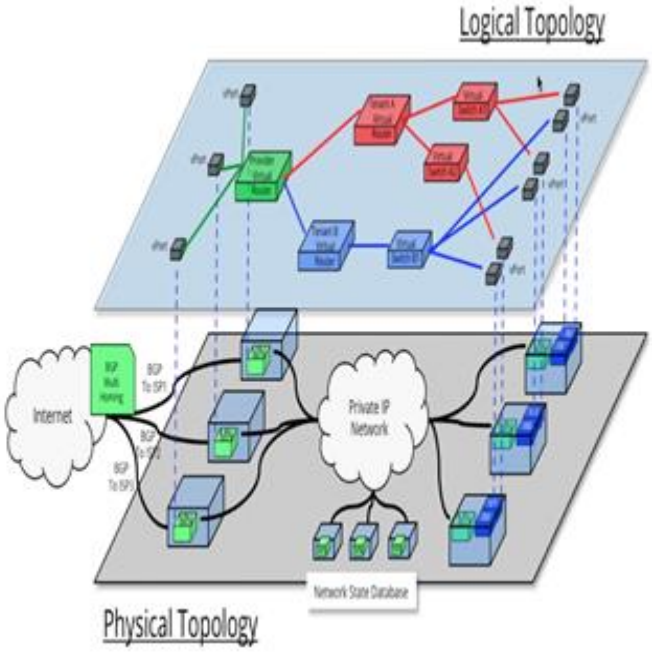


CLOUD COMPUTING COURSE

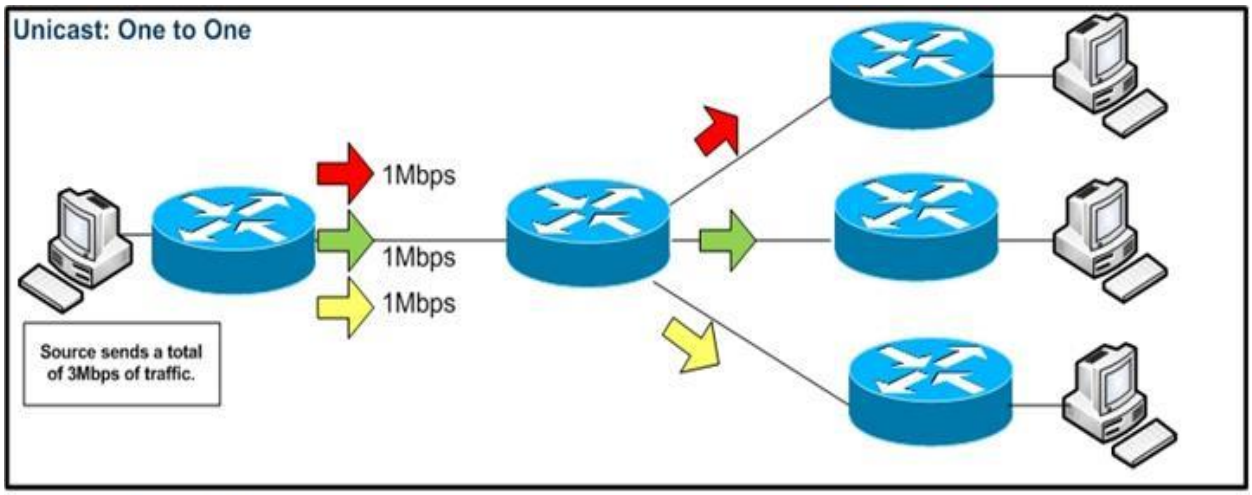
Lecture15-Network Virtualization in CC (4)

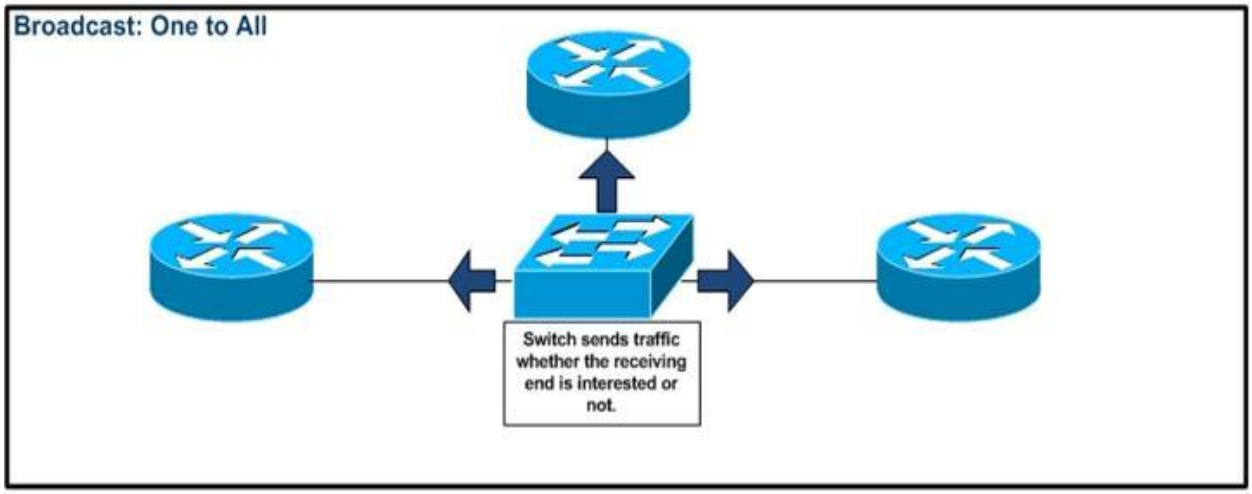
ABSTRACT

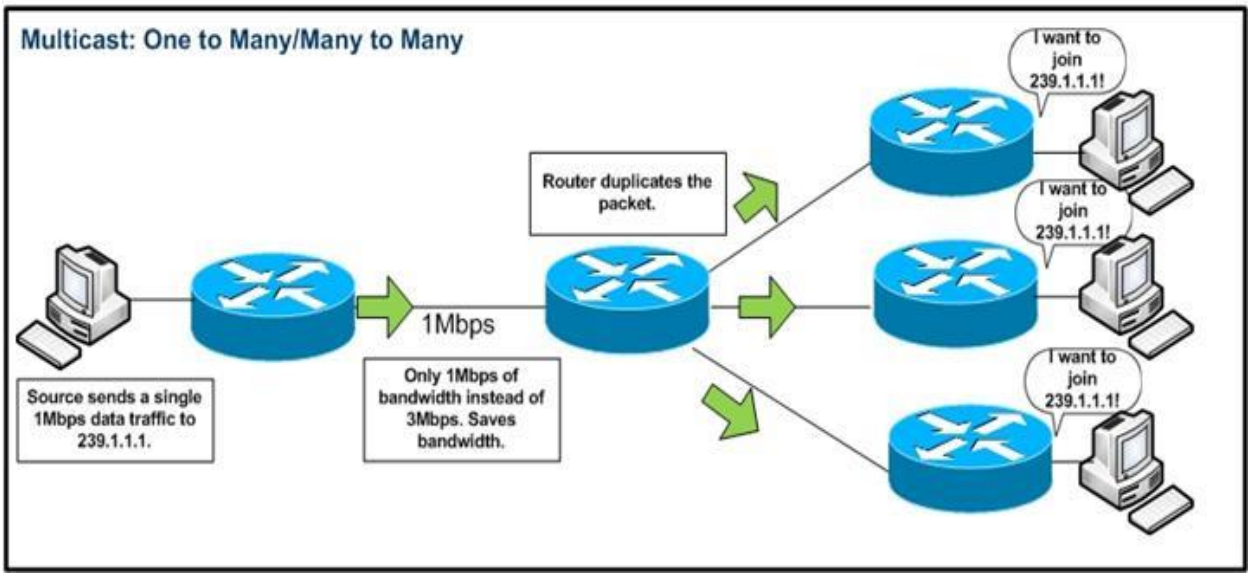
Lecture15 Introduces network traffics and NAT

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1. Network Traffic Delivery Methods







Data is transported over a network by three simple methods:

Unicast, Broadcast, and Multicast.

1. **Unicast**: from one source to one destination i.e. One-to-One

Traffic Example

Many streams of IP packets that move across networks flow from a single point, such as a website server, to a single endpoint such as a client PC. This is the most common form of information transference on networks.

2. **Broadcast:** from one source to all possible destinations i.e. One-to-All

Traffic Example

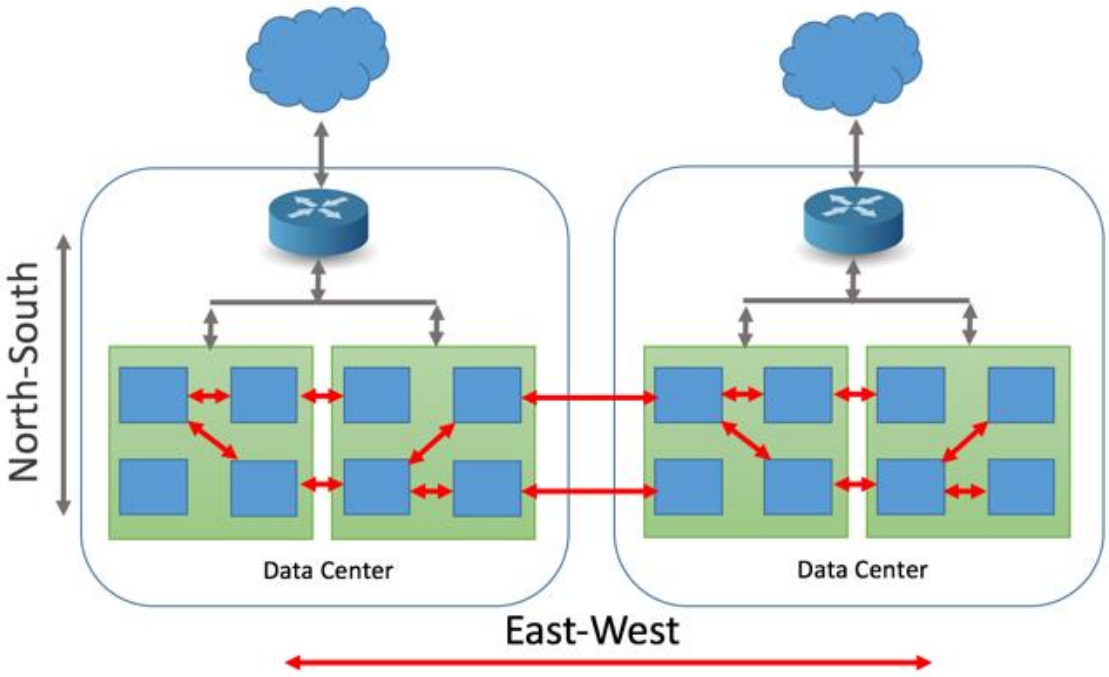
Traffic streams from a single point to all possible endpoints within reach on the network, which is generally a LAN. This is the easiest technique to ensure traffic reaches to its destinations. This mode is mainly utilized by television networks for video and audio distribution.

3. **Multicast:** from one source to multiple destinations stating an interest in receiving the traffic i.e. One-to-Many

Traffic Example

Multicast means that only the destinations that openly point to their requisite to accept the data from a specific source to receive the traffic stream. Multicast routers replicate packets received on one input interface and send the replicas out on multiple output interfaces.

2. Network Traffic Flow



NORTH-SOUTH and **EAST-WEST** traffic is a network traffic flow pattern in the context of a data center. Let us consider the following example to understand these terms.

Example

- Let us say you are trying to access some web application via the browser. The web application is deployed in an application server, which sits in some data center. In a multi-tier architecture, a typical data center will not only contain the application server but also contains other servers such as load balancer, database etc. along with the network components such as routers and switches. Let us say, the application server is front-ended by a load balancer.
- When you access the web application, the following kinds of network flows happen:

1. The network flow between the client (browser that sits out side of a data center) and the load balancer (that sits in a data center)
2. The network flow among the load balancer, application server, data base etc. which all sit with in a data center.

- **NORTH-SOUTH Traffic**

  In the above example, the first kind of network flow (i.e) the network flow between the client and the server is called the NORTH-SOUTH traffic. To put it simply, **NORTH-SOUTH** traffic is a server-client traffic (**external traffic**).

- **EAST-WEST Traffic**

  In the above example, the second kind of network flow (i.e) the network flow among different servers with in a data center (or) the network flow among different data centers themselves is called EAST-WEST traffic. To put it simply, **EAST-WEST** traffic is a **server-server** traffic (**internal traffic**).

  In *Cloud Computing* Systems, the **EAST-WEST** traffic is much more than the NORTH-SOUTH traffic.

Why they are named so?

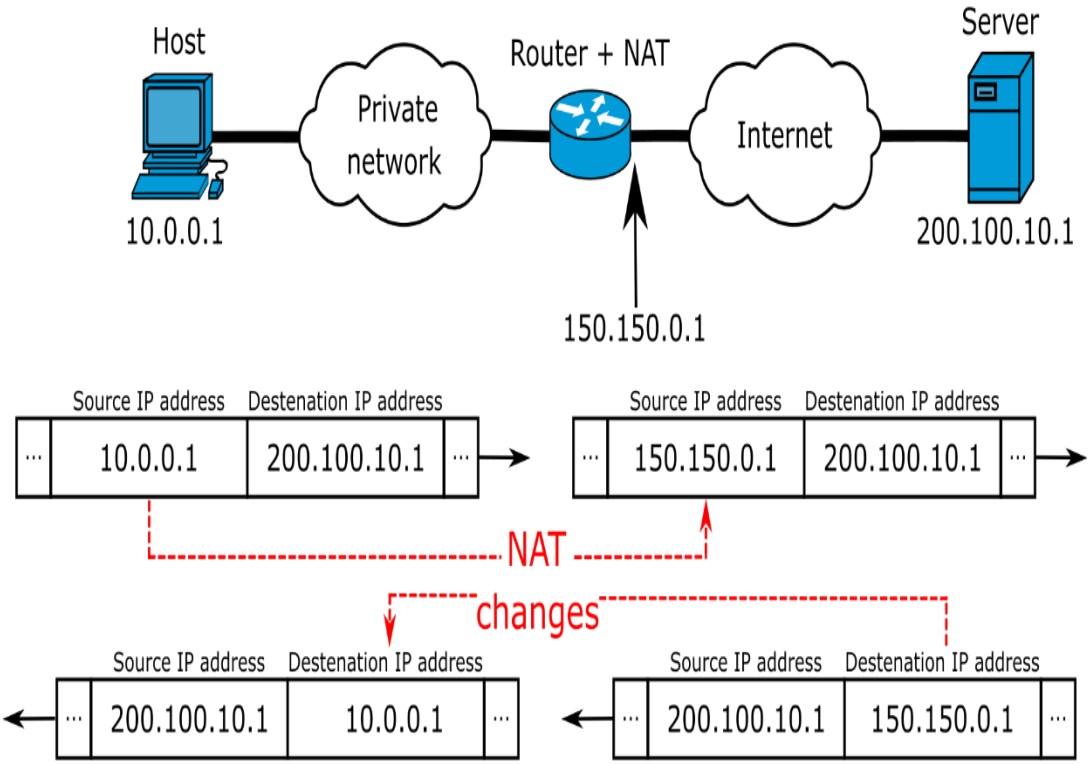
The naming comes from the way the typical network diagrams are drawn to represent the network flow in and out of data center. In diagrams, usually the core network components are drawn at the top (NORTH) and the clients are drawn at the bottom (SOUTH). The different servers with in a data center are drawn in between horizontally (EAST-WEST).

3. NAT and IP Addresses

A NAT (Network Address Translation or Network Address Translator) is the virtualization of Internet Protocol (IP) addresses.

NAT gateways sit between two networks, the inside network and the outside network. Systems on the inside network are typically assigned IP addresses **(Private IP Addresses)** that cannot be routed to external networks. A few externally valid IP addresses **(Public IP Address)** are assigned to the gateway (router/firewall). The gateway makes outbound traffic from an inside system appear to be coming from one of the valid external addresses. It takes incoming traffic aimed at a valid external address and sends it to the correct internal system. This helps ensure security, since each outgoing or incoming request must go through a translation process (NAT Table).

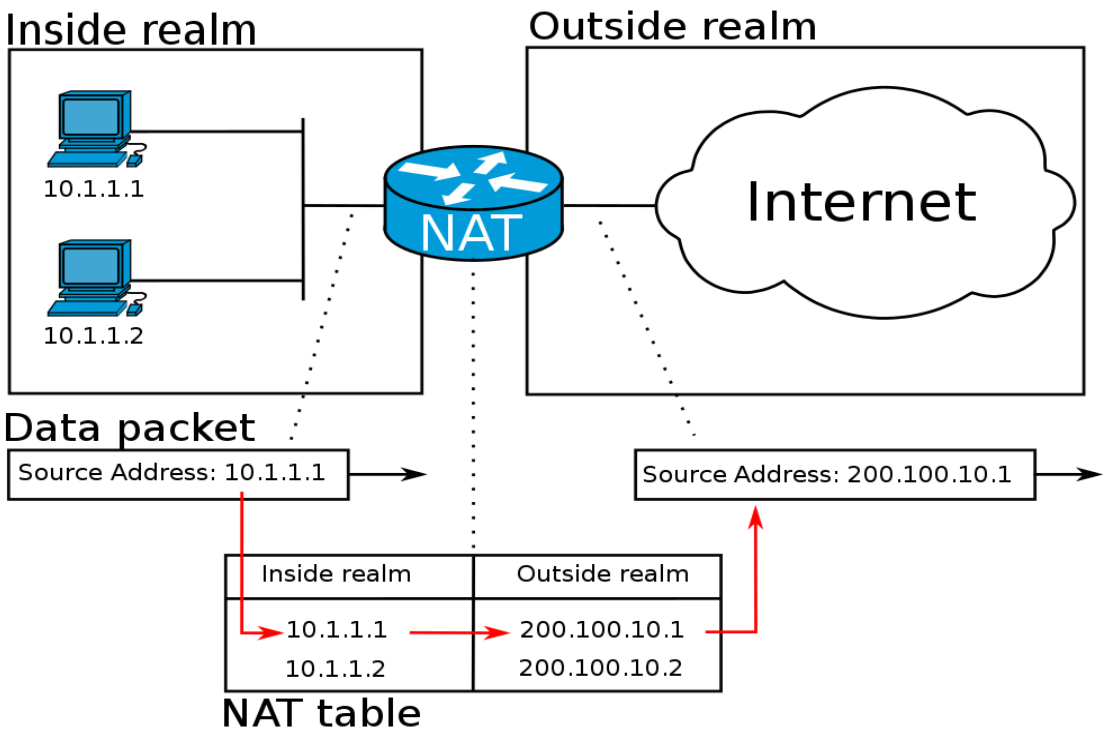
The NAT mechanism ("natting") is a **router** feature, and is often part of a corporate **firewall**.



A router that involves re-writing the source and/or destination IP addresses and usually also the TCP/UDP port numbers of IP packets as they pass through. Checksums must also be rewritten to take account of the changes.

Why NAT?

- Most systems using NAT do so in order to enable multiple hosts on a private network to access the Internet using a single public IP address.
- NAT helps improve security and decrease the number of IP addresses an organization needs.



Please watch NAT Video

https://youtu.be/FTUV0t6JaDA

4. **A gateway** is a networking device that connects two networks using different protocols together. It also acts as a "gate" between two networks. It may be a router, firewall, server, or other devices that enables traffic to flow in and out of the network.

Please watch Firewall Video

https://youtu.be/kDEX1HXybrU